THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 11: **Keeping Time with the Skies** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The apparent daily motion of the Sun from east to west is caused by the: 1

(a) rotation of the Earth on its axis **(b)** revolution of the Earth around the Sun
(c) motion of the Moon **(d)** rotation of the Sun

2. The Moon takes about ___ to go around the Earth once. 1

(a) 27 days **(b)** 365 days **(c)** 24 hours **(d)** 7 days

3. The calendar based on the motion of the Earth around the Sun is the: 1

(a) lunar calendar **(b)** solar calendar
(c) weekly calendar **(d)** lunisolar calendar

4. The Earth completes one rotation on its axis in about: 1

(a) 27 days **(b)** 24 hours **(c)** 365 days **(d)** 12 hours

5. The time from one new moon to the next new moon is about: 1

(a) 365 days **(b)** 7 days **(c)** 24 hours **(d)** 29.5 days

6. The calendar based on the phases of the Moon is the: 1

(a) weekly calendar **(b)** lunar calendar **(c)** solar calendar **(d)** daily

7. The Earth completes one revolution around the Sun in about: 1

(a) 30 days **(b)** 24 hours
(c) 27 days **(d)** 365 and a quarter days

8. A moon phase when the full face of the Moon is visible is the: 1

(a) first quarter **(b)** full moon **(c)** new moon **(d)** crescent

9. A year of the Gregorian calendar has: 1

(a) 360 days **(b)** 400 days
(c) 365 days (366 in a leap year) **(d)** 354 days

10. A leap year has: 1

(a) 360 days **(b)** 365 days **(c)** 364 days **(d)** 366 days

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the Sun appear to rise in the east and set in the west? 2

12. Why is the shadow of a stick shortest at noon? 2

13. Name the two types of calendars based on the Sun and on the Moon. 2

14. What causes day and night? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an activity to make a simple sundial and use it to measure time. 3

16. Explain how the seasons are caused. 3

17. Explain what is meant by 'a day' as the time for one rotation, and a 'month' and a 'year' with astronomical references. 3

OR

Explain how the length and direction of the shadow of a stick change during the day. 3

18. What are the effects of the Earth's rotation? List three. 3

OR

Describe the Jantar Mantar in Jaipur and its use. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe how the Earth's rotation and revolution lead to day and night, years and seasons. 5

OR

Explain why we see the Moon from Earth in phases and not always the full disc, using the positions of the Sun, Earth and Moon. 5

20. Describe the phases of the Moon through a month with reasons and the names for waxing and waning phases. 5

OR

Describe the working of a sundial and the Jantar Mantar's Samrat Yantra. What does each measure? 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND